Routing, capacity and width in dynamic convolution: MobileNetV2

Replication of the 132-run ResNet-18 program

Ankit Kumar
Mechatronics Department, University of Agder, Grimstad, Norway

Generated from result files by `report/build_model_report.py`

SYNTHETIC TEST DATA. Every number in this document is generated to test the pipeline. None is a result.

This document reports one backbone of the cross-architecture replication. It is generated from the result files: every number, table and figure is computed, and every sentence states what a test returned. Interpretation across backbones is in the cross-model report. Method, statistical protocol and the pre-registered margin are those of the ResNet-18 study and are not repeated here.

1 Setup and coverage

Backbone. MobileNetV2, adapted to 32×32 and 64×64 inputs as recorded in `src/models`. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$. Dynamic layers follow the ResNet-18 rule literally (every $k \times k$ convolution with $k > 1$ inside the blocks; stem, 1×1 convolutions, shortcuts, downsampling layers and head static): here the 3×3 depthwise convolution of every inverted-residual block, 17 layers. Initialisation: Kaiming-normal (fan-out) for every static convolution and every bank kernel.

Size. At CIFAR-100 width 64 the static network stores 2.35 M parameters and the $K = 4$ `gap` network 3.68 M, of which 1137.3 K are router parameters (Table 14). The bank adds 0.19 M to the static network; the router is 48.4 % of the static network's size. The router is the ResNet-18 design (a two-layer MLP with hidden width $C_{\text{in}}/4$ on the global descriptor), so its size grows with the square of a layer's input channels while a depthwise kernel grows linearly: where dynamic layers are depthwise, the routed arms carry many more parameters than the bank. The `gap−uniform` routing term includes them.

Gate. The first run, static, CIFAR-100, width 64, seed 0, reached 73.78 % against a pre-registered floor of 65.0 %: passed.

Coverage. 130 of 132 runs are present; 2 are missing, of which 1 diverged (non-finite training loss). 0 group(s) ran with gradient accumulation because batch 128 did not fit in memory. Table 1 lists every missing run and every deviation. Paired tests use the seeds both arms have; a verdict computed on fewer than six pairs carries its n.

Table 1: Missing runs, divergences and deviations from the recipe.

dataset	width	K	τ	arm	seed	status
CIFAR-100	16	4	anneal	`gap`	4	not run
Tiny-ImageNet	64	4	const	`uniform`	5	diverged (non-finite loss)

2 CIFAR-100 at the standard width

Against `static`: `uniform` +0.156 pp (equivalent); `gap` +0.816 pp (higher); `shuffled` +0.257 pp (higher); `frozen_random` +0.242 pp (equivalent). 2 of the four arms are equivalent to `static` within 0.5 pp. The largest gap between arm means is 0.816 pp against a median within-arm seed standard deviation of 0.137 pp; one-way ANOVA across the five arms $F = 23.744$, $p = 0.000$. `frozen_random−gap` $= -0.574$ pp ($p = 0.008$, $p_{\text{TOST}} = 0.6988$). Power for `gap−uniform` here: paired sd 0.267 pp, minimum detectable difference 0.281 pp, simulated power 45.7, 95.1 and 100.0 % at true effects of 0.25, 0.5 and 1.0 pp. Table 2 gives the routing entropy by arm.

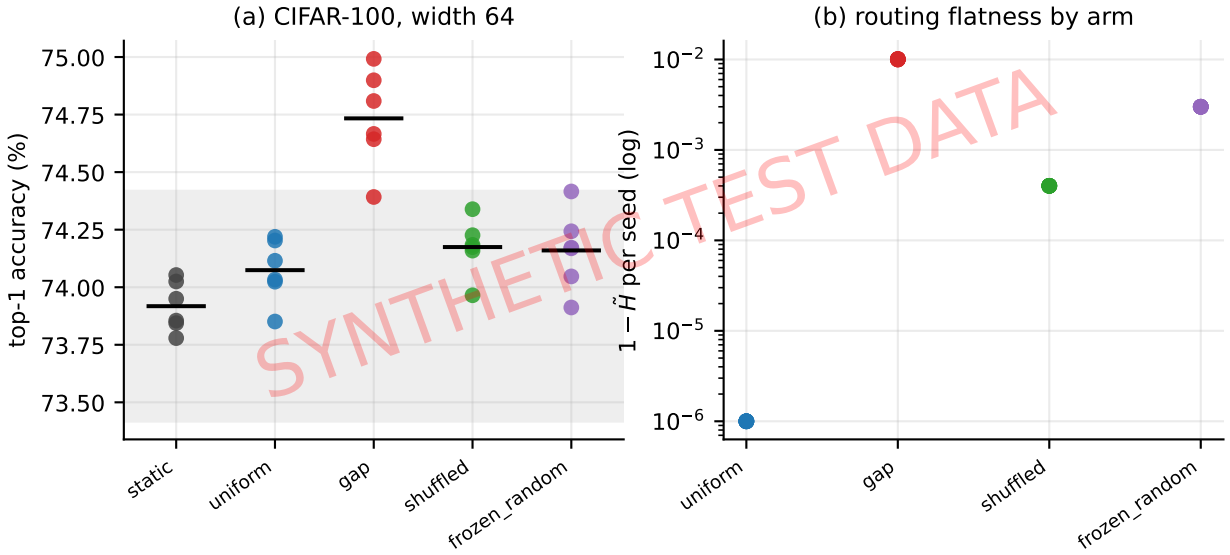


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

Table 2: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9899	0.9898	0.9900	0.01005
`shuffled`	0.9996	0.9996	0.9996	0.00040
`frozen_random`	0.9970	0.9970	0.9970	0.00300

3 The capacity sweep on CIFAR-100

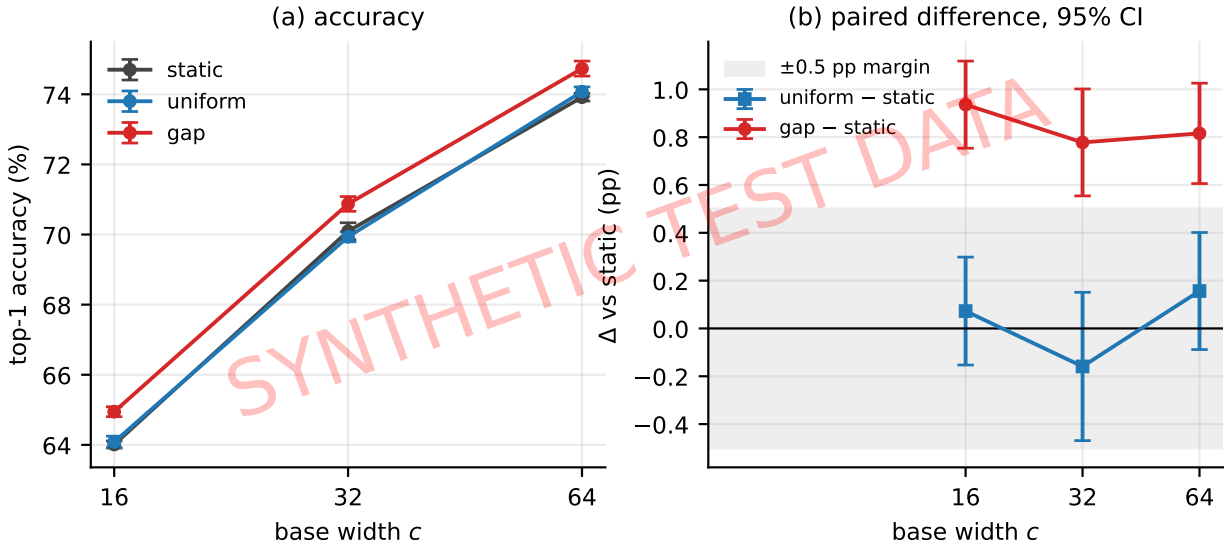


Figure 2: CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference against `static` with 95 % confidence intervals; the band is the ±0.5 pp margin.

`gap−static`: +0.936 pp at width 16 (higher), +0.778 pp at width 32 (higher), +0.816 pp at width 64 (higher) (Figure 2, Table 13).

4 Bank against routing

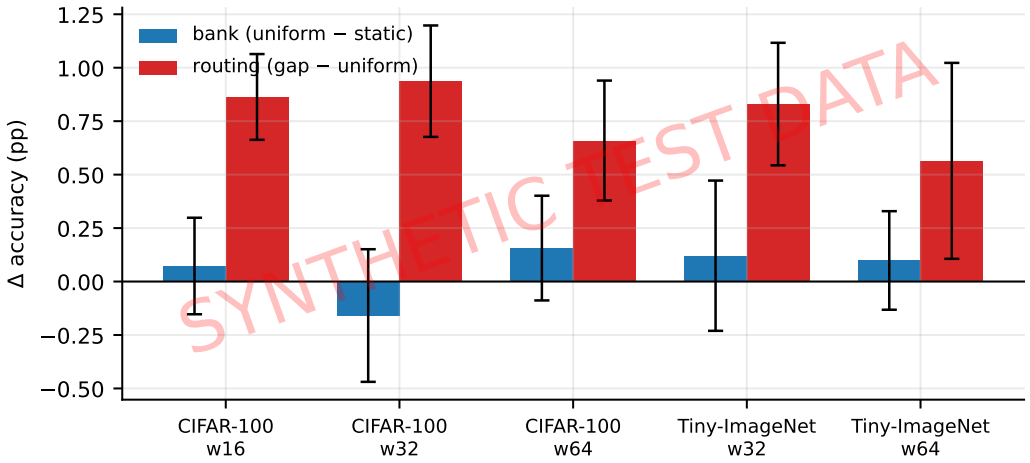


Figure 3: The total effect split into a bank term (`uniform−static`) and a routing term (`gap−uniform`), five conditions, 95 % confidence intervals.

The bank term is equivalent to zero in 5 of 5 conditions; the others: none. The routing term is positive with $p < 0.05$ in 5 of 5 conditions: CIFAR-100 width 16 +0.863 pp ($p = 0.0001$); CIFAR-100 width 32 +0.937 pp ($p = 0.0002$); CIFAR-100 width 64 +0.659 pp ($p = 0.0018$); Tiny-ImageNet width 32 +0.830 pp ($p = 0.0007$); Tiny-ImageNet width 64 +0.564 pp ($p = 0.0268$) (Table 3, Figure 3).

Table 3: Decomposition. Bank verdicts use TOST at ±0.5 pp.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	+0.073	0.0023	equivalent	+0.863	0.0001	+0.936	$< 10^{-4}$
CIFAR-100	32	-0.159	0.0184	equivalent	+0.937	0.0002	+0.778	0.0003
CIFAR-100	64	+0.156	0.0077	equivalent	+0.659	0.0018	+0.816	0.0002
Tiny-ImageNet	32	+0.121	0.0196	equivalent	+0.830	0.0007	+0.951	0.0059
Tiny-ImageNet	64	+0.099	0.0042	equivalent (n=5)	+0.564	0.0268	+0.744	0.0063

5 Bank size

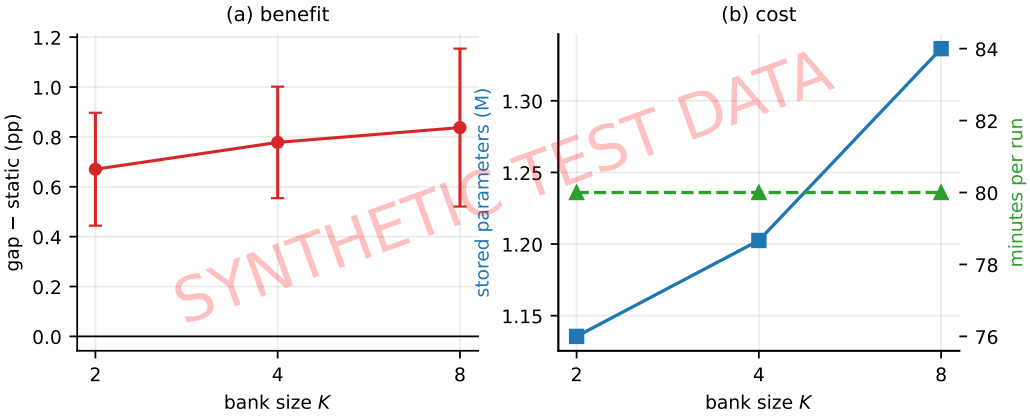


Figure 4: CIFAR-100 width 32, `gap` with $K = 2, 4, 8$. (a) Difference against `static`, 95 % CI. (b) Stored parameters and minutes per run.

$K = 8$ against $K = 2$: +0.167 pp (equivalent) for 1.18× the parameters (Tables 4, 5).

Table 4: Bank size at CIFAR-100 width 32. Differences are against `static`.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	70.765	0.151	+0.670	[+0.444, +0.897]	0.0006	1.14	80.0
4	70.873	0.210	+0.778	[+0.554, +1.002]	0.0003	1.20	80.0
8	70.932	0.085	+0.837	[+0.521, +1.154]	0.0010	1.34	80.0
`static`	70.095	0.240	–	–	–	0.82	20.0

Table 5: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.108	[-0.216, +0.431]	0.4316	0.0132	equivalent
$K = 8$ vs $K = 4$	+0.059	[-0.209, +0.328]	0.5942	0.0042	equivalent
$K = 8$ vs $K = 2$	+0.167	[-0.031, +0.365]	0.0820	0.0038	equivalent

6 Tiny-ImageNet

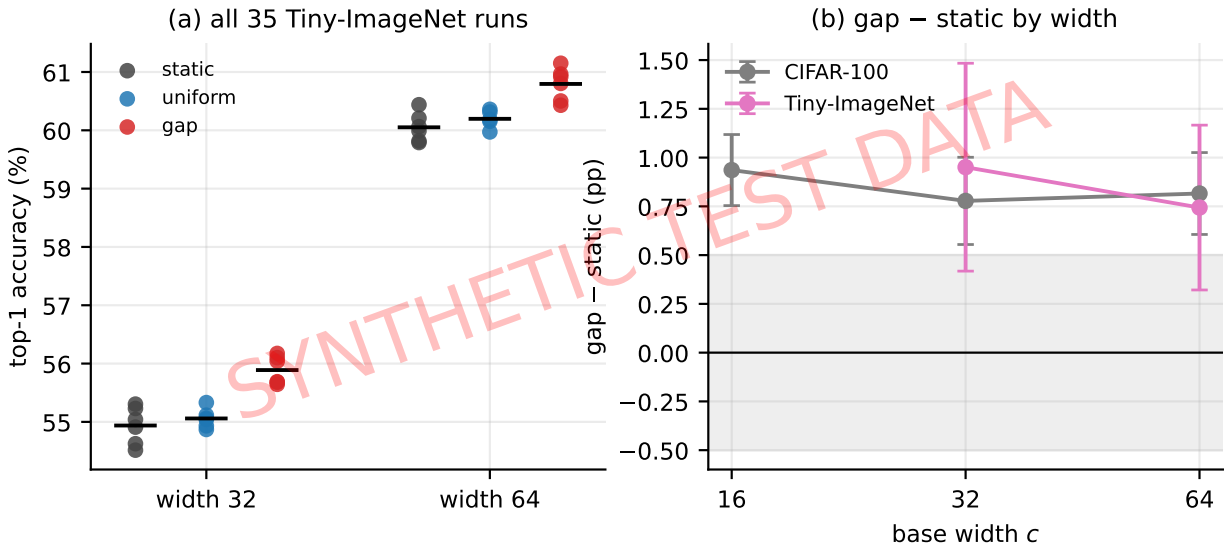


Figure 5: Tiny-ImageNet. (a) Every run at widths 32 and 64. (b) `gap−static` against width on both datasets, 95 % CI; the band is the margin.

`gap−static` is +0.951 pp at width 32 (higher) and +0.744 pp at width 64 (higher).

Across the five conditions the Spearman correlation between the static arm's final training loss and `gap−static` is $\rho = +0.60$ (Table 6, Figure 6).

Table 6: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap − static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0244	+0.816	[+0.606, +1.026]	higher
Tiny-ImageNet	64	0.0306	+0.744	[+0.321, +1.166]	higher
CIFAR-100	32	0.0619	+0.778	[+0.554, +1.002]	higher
Tiny-ImageNet	32	0.0869	+0.951	[+0.418, +1.484]	higher
CIFAR-100	16	0.1119	+0.936	[+0.754, +1.118]	higher

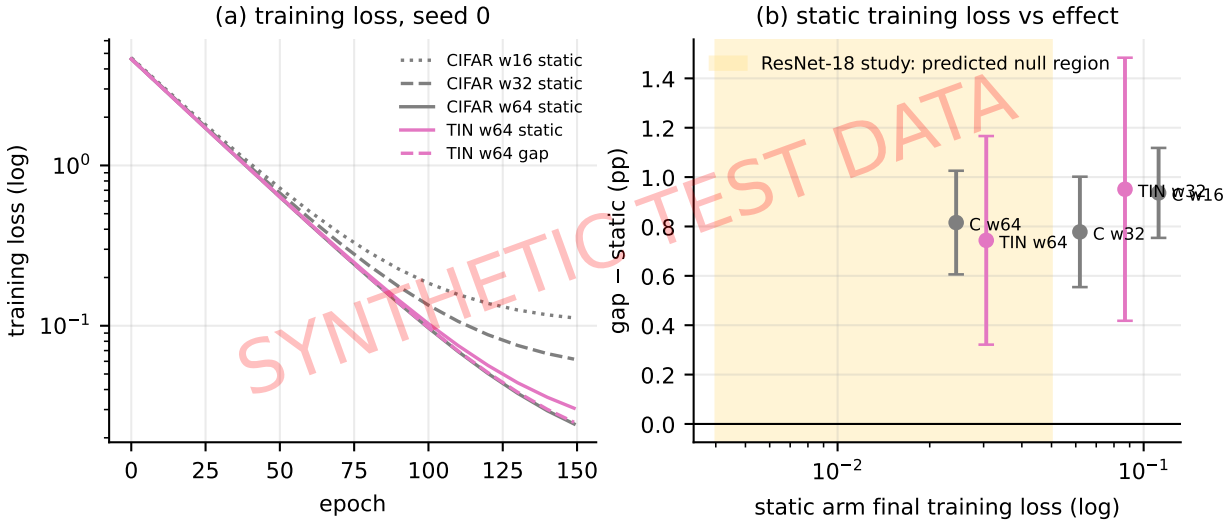


Figure 6: (a) Training loss, seed 0, sampled every ten epochs. (b) `gap−static` against the static arm's final training loss; the shaded region is where the ResNet-18 study predicted a null.

7 Equal parameter budgets

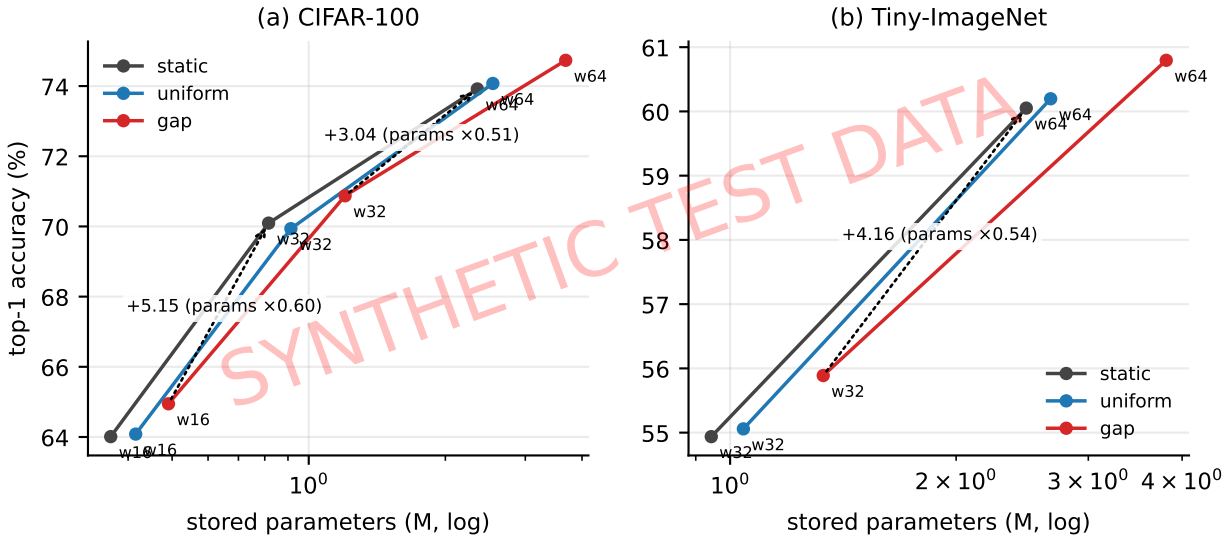


Figure 7: Accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; the label is the accuracy difference, and the parameter ratio where it is outside $\pm 10\,\%$.

The ResNet-18 comparison of `gap` at width c with `static` at width $2c$ was a matched-budget comparison because a $K = 4$ bank and a doubling of width both multiply the parameters of the dynamic layers by four. That holds only where almost every parameter is in a dynamic layer. Here 0 of 3 pairs are within 10 % and are called matched: CIFAR-100 `gap` w16 vs `static` w32: +5.149 pp, parameter ratio 0.60 (not matched); CIFAR-100 `gap` w32 vs `static` w64: +3.045 pp, parameter ratio 0.51 (not matched); Tiny-ImageNet `gap` w32 vs `static` w64: +4.163 pp, parameter ratio 0.54 (not matched) (Table 7). Table 8 adds the comparison at equal parameters: the static accuracy curve interpolated linearly in log-parameters between the static widths that were run, evaluated at the `gap` network's count. It is an interpolation, not a run, and has no confidence interval.

Table 7: `gap` at width c against `static` at width $2c$. Δ is plain minus routed. The last column is the routed network's parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (0.49M)	`static` w32 (0.82M)	64.946	70.095	+5.149	[+4.815, +5.483]	$< 10^{-4}$	40.0 / 20.0	0.60 (not matched)
CIFAR-100	`gap` w32 (1.20M)	`static` w64 (2.35M)	70.873	73.918	+3.045	[+2.812, +3.277]	$< 10^{-4}$	80.0 / 40.0	0.51 (not matched)
Tiny-ImageNet	`gap` w32 (1.33M)	`static` w64 (2.48M)	55.889	60.051	+4.163	[+3.822, +4.504]	$< 10^{-4}$	80.0 / 40.0	0.54 (not matched)

Table 8: `gap` against the static accuracy curve at equal parameters (interpolated, no CI).

dataset	routed	params (M)	routed acc.	static at equal params	Δ (pp)
CIFAR-100	`gap` w16	0.49	64.946	66.237	+1.291
CIFAR-100	`gap` w32	1.20	70.873	71.496	+0.623
Tiny-ImageNet	`gap` w32	1.33	55.889	56.755	+0.867

8 Temperature schedule

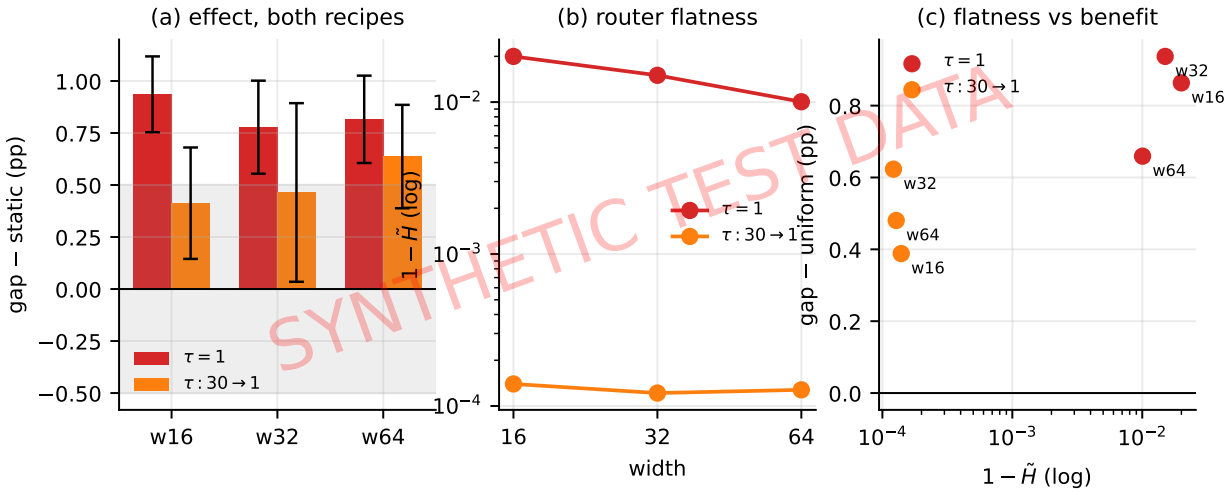


Figure 8: Constant $\tau = 1$ against τ annealed from 30 to 1 over ten epochs, CIFAR-100. (a) `gap−static`. (b) Router flatness. (c) Flatness against benefit over `uniform`.

Annealed minus constant, `gap`: width 16 -0.509 pp (lower (n=5)); width 32 -0.314 pp (inconclusive); width 64 -0.179 pp (equivalent) (Table 9).

Table 9: Temperature schedule, CIFAR-100.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	64.946	64.441	-0.509	0.0110	+0.936	+0.413	0.2081	+0.388	inconclusive (n=5)
32	70.873	70.559	-0.314	0.0723	+0.778	+0.464	0.4193	+0.623	higher
64	74.734	74.555	-0.179	0.1592	+0.816	+0.637	0.8919	+0.480	higher

9 Routing entropy and routing benefit

At width 16 the constant-temperature router's $1 - \tilde{H}$ is 143.1 times the annealed router's; the benefit over `uniform` is +0.863 pp constant and +0.388 pp annealed. Across the 6 width–recipe points the Spearman correlation between $1 - \tilde{H}$ and `gap−uniform` is $\rho = +0.71$ (Table 10).

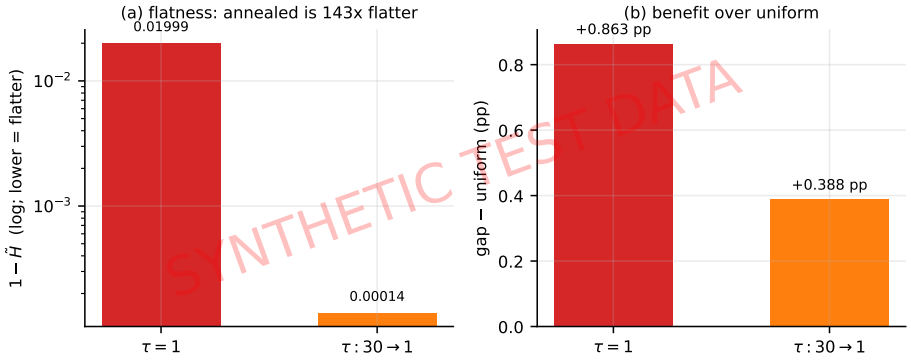


Figure 9: CIFAR-100 width 16, both temperature recipes. (a) Router flatness. (b) Benefit of `gap` over `uniform`.

Table 10: Routing entropy against the benefit over `uniform`.

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap`−`uniform` (pp)	p
16	1	0.98001	0.01999	0.0588	+0.863	0.0001
16	$30 \to 1$	0.99986	0.00014	0.0049	+0.388	0.0776
32	1	0.98499	0.01501	0.0510	+0.937	0.0002
32	$30 \to 1$	0.99988	0.00012	0.0046	+0.623	0.0004
64	1	0.98995	0.01005	0.0417	+0.659	0.0018
64	$30 \to 1$	0.99987	0.00013	0.0047	+0.480	$< 10^{-4}$

10 Pre-registered predictions

Recorded in `PREREGISTRATION_MODELS.md` on 30 September 2026, before any run of this backbone.

Table 11: Pre-registered predictions and their outcomes.

	prediction	measured	outcome
P1	bank term equivalent to zero in each of 5 conditions	5 of 5	held
P2	static at width $2c$ beats `gap` at width c in each of 3 pairs	3 of 3	held
P3	routing term positive, $p < 0.05$, in at least 3 of 5 conditions	5 of 5	held

11 Complete results

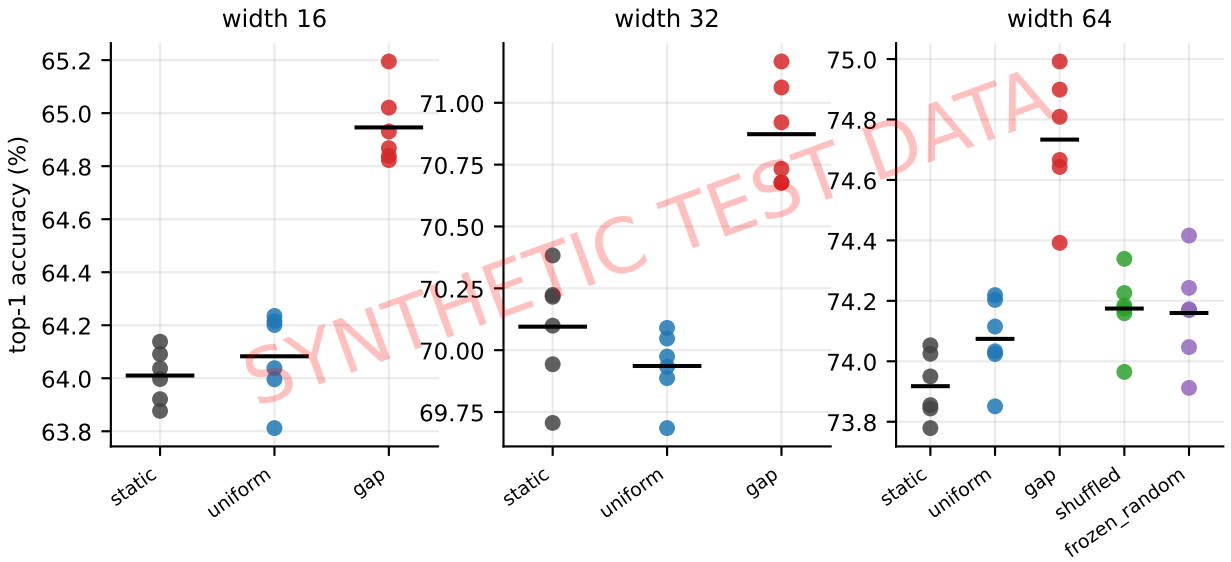


Figure 10: Every CIFAR-100 run at $K = 4$, $\tau = 1$. Bars are arm means.

Table 12: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). – marks a missing or diverged run (Table 1).

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	64.14	64.09	64.00	64.04	63.92	63.88	64.010	0.099
CIFAR-100	16	`uniform`	64.22	64.04	64.00	63.81	64.23	64.20	64.083	0.166
CIFAR-100	16	`gap`	64.87	65.19	64.84	64.82	64.93	65.02	64.946	0.142
CIFAR-100	32	`static`	70.22	70.10	69.94	70.38	70.22	69.71	70.095	0.240
CIFAR-100	32	`uniform`	70.05	69.97	69.69	69.89	69.93	70.09	69.936	0.143
CIFAR-100	32	`gap`	70.73	71.06	70.68	70.92	71.17	70.68	70.873	0.210
CIFAR-100	64	`static`	73.78	73.86	73.84	74.05	73.95	74.03	73.918	0.109
CIFAR-100	64	`uniform`	74.22	74.02	74.20	73.85	74.11	74.03	74.074	0.137
CIFAR-100	64	`gap`	74.81	74.64	74.39	74.67	74.90	74.99	74.734	0.214
CIFAR-100	64	`shuffled`	74.18	74.17	74.23	73.97	74.16	74.34	74.174	0.122
CIFAR-100	64	`frozen_random`	74.42	74.24	74.17	74.17	74.05	73.91	74.160	0.171
Tiny-ImageNet	32	`static`	55.30	54.63	55.23	54.52	55.04	54.91	54.938	0.319
Tiny-ImageNet	32	`uniform`	55.33	55.12	54.87	55.04	55.06	54.93	55.059	0.161
Tiny-ImageNet	32	`gap`	55.68	56.10	55.64	56.04	56.17	55.69	55.889	0.242
Tiny-ImageNet	64	`static`	59.79	60.21	60.44	59.99	60.06	59.82	60.051	0.246
Tiny-ImageNet	64	`uniform`	60.15	60.36	60.31	59.97	60.19	–	60.197	0.152
Tiny-ImageNet	64	`gap`	60.43	60.80	60.50	60.92	61.15	60.97	60.795	0.279

Table 13: Paired comparisons against `static`. p_W is the Wilcoxon signed-rank p (floor 0.031 at $n = 6$).

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	+0.073	[-0.153, +0.298]	0.4456	0.5625	+0.34	0.0023	equivalent
CIFAR-100	16	`gap`	+0.936	[+0.754, +1.118]	$< 10^{-4}$	0.0312	+5.39	0.9992	higher
CIFAR-100	32	`uniform`	-0.159	[-0.469, +0.151]	0.2447	0.3125	-0.54	0.0184	equivalent
CIFAR-100	32	`gap`	+0.778	[+0.554, +1.002]	0.0003	0.0312	+3.65	0.9879	higher
CIFAR-100	64	`uniform`	+0.156	[-0.088, +0.401]	0.1613	0.2188	+0.67	0.0077	equivalent
CIFAR-100	64	`gap`	+0.816	[+0.606, +1.026]	0.0002	0.0312	+4.08	0.9941	higher
CIFAR-100	64	`shuffled`	+0.257	[+0.066, +0.448]	0.0182	0.0625	+1.41	0.0111	higher
CIFAR-100	64	`frozen_random`	+0.242	[-0.035, +0.519]	0.0743	0.0938	+0.92	0.0310	equivalent
Tiny-ImageNet	32	`uniform`	+0.121	[-0.231, +0.472]	0.4171	0.2188	+0.36	0.0196	equivalent
Tiny-ImageNet	32	`gap`	+0.951	[+0.418, +1.484]	0.0059	0.0312	+1.87	0.9592	higher
Tiny-ImageNet	64	`uniform`	+0.099	[-0.132, +0.329]	0.3006	0.4375	+0.53	0.0042	equivalent (n=5)
Tiny-ImageNet	64	`gap`	+0.744	[+0.321, +1.166]	0.0063	0.0312	+1.85	0.9010	higher

Table 14: Configuration and cost. Router parameters in thousands; training loss is the final-epoch mean over seeds; minutes are logged wall clock.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.37	0.0	0.1119	10.0
CIFAR-100	16	`uniform`	0.42	0.0	0.1119	40.0
CIFAR-100	16	`gap`	0.49	75.3	0.0819	40.0
CIFAR-100	32	`static`	0.82	0.0	0.0619	20.0
CIFAR-100	32	`uniform`	0.91	0.0	0.0619	80.0
CIFAR-100	32	`gap`	1.20	289.1	0.0469	80.0
CIFAR-100	64	`static`	2.35	0.0	0.0244	40.0
CIFAR-100	64	`uniform`	2.54	0.0	0.0244	160.0
CIFAR-100	64	`gap`	3.68	1137.3	0.0206	160.0
CIFAR-100	64	`shuffled`	3.68	1137.3	0.0244	160.0
CIFAR-100	64	`frozen_random`	2.54	0.0	0.0244	160.0
Tiny-ImageNet	32	`static`	0.94	0.0	0.0869	20.0
Tiny-ImageNet	32	`uniform`	1.04	0.0	0.0869	80.0
Tiny-ImageNet	32	`gap`	1.33	289.1	0.0644	80.0
Tiny-ImageNet	64	`static`	2.48	0.0	0.0306	40.0
Tiny-ImageNet	64	`uniform`	2.67	0.0	0.0306	160.0
Tiny-ImageNet	64	`gap`	3.81	1137.3	0.0250	160.0

Table 15: Every run of this backbone. GPU-hours from logged wall clock.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	9.0
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	18.0
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	68.0
CIFAR-100	32	2	constant	`gap`	6	8.0
CIFAR-100	32	8	constant	`gap`	6	8.0
CIFAR-100	16	4	annealed	`gap`	5	3.3
CIFAR-100	32	4	annealed	`gap`	6	8.0
CIFAR-100	64	4	annealed	`gap`	6	16.0
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	18.0
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	17	33.3
total					130	189.7